Spotlight Reflex

Minimal-Shot Autonomous Driving via Grounded ManeuverTokens

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The Problem — Continuous Policy Collapse in Long-Tail

Imitation learning minimises:

$\min_f \; \mathbb{E}_{(x,y)\sim\mathcal{D}}\big[\mathcal{L}(f(x), y)\big]$

Works on the common driving distribution $\mathcal{D}$.

Fails when $P_{\textbf{test}} \neq P_{\textbf{train}}$: continuous distributions spread — the policy assigns probability mass to unsafe intermediate states that never appeared in training.

WOD-E2E targets **11 scenario types** occurring $<$0.03% of driving time: wrong-way actors, animals, debris, construction, cut-ins, cyclists.

Continuous baseline: no world-state reasoning. Drives directly into the actor. Gauntlet pass rate: **2.1%**, collision: **20.5%**.

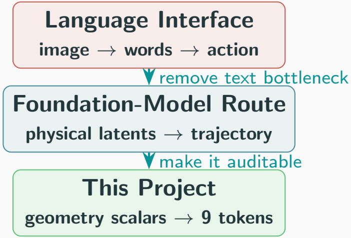
Wrong-way actor, night. Continuous baseline: no awareness the path is occupied. Policy variance under OOD → collision.

Motivation: Grounded Decisions, Not Language Plans

Recent VLA work gives a useful direction: driving decisions need physical grounding, because language is a lossy interface for meter-level geometry, timing, clearance, and acceleration.

Related context: LaST-VLA (arXiv:2603.01928)
Xiaomi EV + Tsinghua align latent states with **world-model dynamics** and **3D geometric priors**. That is a foundation-model path to physical grounding.

Our work is deliberately different: we build a compact testbed where grounding is explicit and auditable. The car reasons through route blockage, clearance, actor motion, and a fixed set of maneuver tokens.



Positioning: LaST-VLA is a large VLA training framework. This project is an experimental autonomy stack: simulator, token runtime, AlpaSim bridge, WOD selector, and failure analysis.

Three Claims

Claim 1 — Discrete Bottleneck Hypothesis
Quantizing the action space to **9 ManeuverTokens** is a structural robustness prior. Fixed geometric semantics prevent policy variance from spreading to unsafe intermediate states.
Evidence: 0 collisions / 350 rollouts; 27× gauntlet pass.

Claim 2 — Geometric Grounding of Visual Latents
Cosmos 64d embeddings carry preference signal inaccessible to linear models but extractable by a GPU MLP. Generation-oriented VLAs fail on discrimination tasks.
Evidence: linear head → no gain; GPU MLP → +0.042 RFS.

Claim 3 — Causal Bottleneck Analysis
ManeuverToken selection is causally determined by geometric pressure scalars — no demonstration data, no spurious correlations. This precisely localizes the 1.419 RFS oracle gap to the visual-to-geometric bridge.
Evidence: failure modes trace to scalar interactions; oracle gap not in action space or ranker.

Architecture: Six World-State Scalars

All signals computed from raw obstacle positions and route geometry. **Nothing is learned; nothing is scene-specific.**

Signal	Definition	Range
`obstacle_pressure`	$p = \mathrm{clip}\left(\frac{10-d_{\mathrm{min}}}{10}, 0, 1\right)$	$[0, 1]$
`route_blockage`	fraction of forward corridor occupied	$[0, 1]$
`corridor_blocked`	hard blockage within stopping distance	bool
`left_clearance`	free lateral space to the left (m)	≥ 0
`right_clearance`	free lateral space to the right (m)	≥ 0
`preferred_escape_side`	$\arg\max(\mathrm{left}, \mathrm{right})$	L / R

d_{min} = distance to nearest obstacle. Pressure = 1.0 at contact, 0 beyond 10 m. No object identity, no semantic labels, no history. All O(1) to compute. Formal invariance tested: $M(s) = M(T(s))$ for any object-identity relabelling T.

Architecture: Nine ManeuverTokens — The Discrete Bottleneck

Each token: **20-point, 5 s trajectory at 4 Hz** via Hermite smoothstep. Evasive maneuvers apply the lateral offset in the first third for fast separation.

Maneuver	Intent	Speed	Lateral
`stop`	Emergency stop	0.00×	0.0 m
`crawl`	Very slow progress	0.15×	0.0 m
`maintain`	Hold current speed	1.00×	0.0 m
`slow_yield`	Decelerate & yield	0.50×	0.0 m
`nudge_left`	Minor left shift	0.80×	+0.9 m
`nudge_right`	Minor right shift	0.80×	−0.9 m
`evasive_left`	Strong left evasion	0.70×	+2.2 m
`evasive_right`	Strong right evasion	0.70×	−2.2 m
`lane_recover`	Return to centre	0.90×	0.0 m

Why discrete?
A continuous policy assigns mass to unsafe *intermediate states* between tokens. The token constraint eliminates those states by construction — uncertainty changes **scores**, not physical footprints.

Coverage: 5 longitudinal × 4 lateral directions covers all safety-critical responses in the Waymo long-tail taxonomy. Nothing missing; nothing extra.

Architecture: Trust-Region Scoring & Decision Flow

Speed-scaled tolerance:

$$s(v) = \mathrm{clip}\left(\frac{v-1.4}{9.6}, 0, 1\right) \times 0.5 + 0.5$$

Horizon	Lateral	Longitudinal
3 s	$1.0 \times s(v)$ m	$4.0 \times s(v)$ m
5 s	$1.8 \times s(v)$ m	$7.2 \times s(v)$ m

$$S(\tau) = 0.5 \cdot \max_r \text{score}_{3s} + 0.5 \cdot \max_r \text{score}_{5s}$$

Scores ≤ 50 = unsafe floor. Clearance < 0.55 m: -1000 penalty.

Decision pipeline (4 steps):

1. Scene $\rightarrow$ compute 6 world-state scalars
2. Reference rule engine maps scalars to expected tokens + scores (76–94)
3. Each of the 9 tokens scored against references at 3 s and 5 s
4. Token with highest combined score is selected

Every step emits a full log: maneuver, scores, decision reason, all scalar values. In AlpaSim: serialised as `reasoning_text` JSON. **Identical policy binary** in 2D simulator and AlpaSim.

Simulation: Gauntlet — The Bottleneck Test

Gauntlet: 4 simultaneous hazards, 3.5 m corridor, 420 matched scenarios (seeds 1–80).

Policy	Pass	Collision
Continuous baseline	2.1%	20.5%
Spotlight Reflex	**57.6%**	**7.9%**
Improvement	*27×*	*2.6× fewer*

Token constraint eliminates unsafe intermediate states. Baseline spreads mass across trajectory space; collision rate: **20.5%**.

evasive_right, clears wrong-way actor, recovers lane centre.

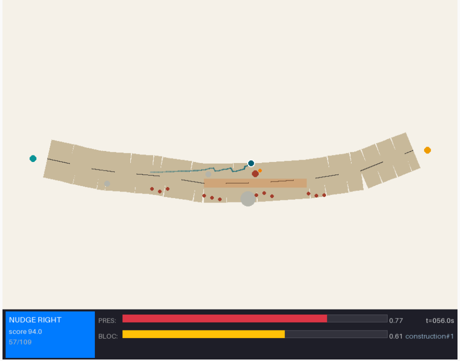
Simulation: 350 Rollouts — Zero Collisions

Primary benchmark — seeds 1–10, deterministic. Same policy binary across all suites.

Suite	Runs	Coll.	Pass
WOD-style (11 clusters)	110	0	100%
Compositional OOD	60	0	100%
Adversarial (2–3 hazards)	60	0	100%
Hidden holdout	60	0	100%
Gauntlet (4h, 3.5 m)	60	0	60%
Total	**350**	**0**	**93.1%**

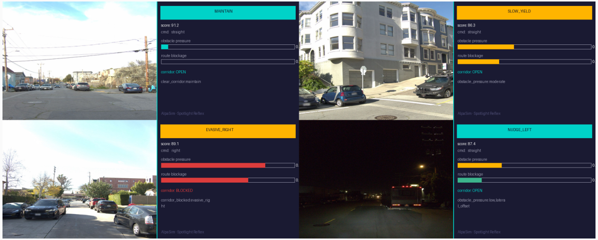
COMPASS **9.137/10** over 700 ranked runs
95% CI collision rate: **[0.0, 0.0053]**
Mean minimum clearance: 2.96 m

OOD robustness: compositional generator samples topology × hazard × weather independently. 8 topologies × 11 hazards × 4 weather modes.



Construction zone — 5.2 m corridor, cone field, lane closure. Token: `nudge_right`*. Policy selects from geometry, not scenario label.*

AlpaSim: What We Built



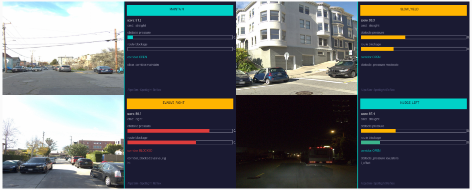
Real Waymo front-camera frames alongside adapter output. The policy API is fixed; the adapter reconstructs proxy geometry.

Engineering work completed:

- Local AlpaSim plugin wrapping the same token policy API
- Front-camera + route + ego-dynamics adapter
- Proxy-state reconstruction into Spotlight/DAgger features
- Docker/runtime bootstrap scripts
- Scene presets and paired matrix runner
- Per-frame token, adapter, collision, offroad, lane, progress logs

This made the simulator result auditable in a second execution stack, instead of staying as a closed internal demo.

AlpaSim: External Transfer Matrix



Same AlpaSim bridge, now used to run paired learned-policy transfer matrices over shared WOD-E2E clips.

10 shared WOD-E2E clips, paired by scene:

Variant	Coll.	Offroad	Lane	Prog.
Raw DAgger iter2	0.60	0.90	0.70	0.034
Clamped DAgger	0.70	**0.50**	**0.20**	0.384
Hard-veto hybrid	0.80	**0.20**	0.70	**0.837**
Source decay	0.60	0.90	0.40	0.175

Interpretation: each intervention fixes a different axis. Clamping improves route geometry; hard-veto recovers progress/offroad; source decay helps lane behavior. No single learned variant dominates all axes.

This is the transfer finding: internal success is not enough unless the selector is grounded at execution time.

WOD-E2E Benchmark: Trajectory Selector

479 preference-labeled Waymo frames · segment-grouped 5-fold CV · RFS = preference alignment

Selector	Folds	RFS
Official Waymo baseline	—	7.022
Local CV baseline	—	7.131
Gate only (HGB)	5	7.803
RFF direct policy	5	7.834
GPU MLP + Cosmos 64d	**5**	**7.845**
HGB Optuna peak†	2	7.880
Oracle (perfect)	5	**9.264**

† 2-fold; not comparable to 5-fold results.

Pipeline: kinematic + ridge + temporal candidates → HGB gate (137 features) → GPU MLP override (4.2% of frames, precision 0.60).

Visual embedding results:

Config	Head	RFS
No embedding	Ridge	7.803
Cosmos 64d	Ridge	≈7.803
InternVLA 128d	Ridge	≈7.803
RFF gate	non-lin.	7.834
Cosmos 64d	**GPU MLP**	**7.845**

Cosmos embeddings carry signal — only a non-linear model extracts it. InternVLA fails: generation $\neq$ discrimination.

GPU MLP — h=64, lr=0.003866, batch=512, 10 epochs, Optuna TPE.

Oracle Gap: Causal Localization of the Bottleneck

Oracle gap: 1.419 RFS (9.264 − 7.845)

The right candidate exists in the pool. Where does the gap live?

Slice	Sel.	Regret
GO_STRAIGHT (n=427)	7.959	1.377
GO_LEFT (n=23)	7.107	**1.614**
GO_RIGHT (n=29)	6.574	**2.063**
speed:urban (n=136)	8.132	1.285
speed:slow (n=133)	7.564	**1.632**
speed:fast (n=44)	7.957	1.479

Causal elimination:

- **Not the action space** — oracle 9.264; preferred candidate exists in pool.
- **Not the ranker** — gate 7.803 + GPU MLP 7.845 exhaust trajectory + visual feature information.
- **The missing bridge:** model that infers geometric context from camera would resolve the ∼15% of frames where gate and direct policy both fail.

GO_RIGHT (2.063 regret, n=29): temporal gate over-selects. Rightward intent requires lateral clearance checks absent from ego-history — a geometry-aware camera model would catch this.

Three Contributions — Backed by Evidence

Contribution	Evidence
1. Discrete Bottleneck	27× gauntlet pass (57.6% vs 2.1%). 0 collisions / 350 rollouts. COMPASS 9.137/10. Collision: 7.9% vs 20.5% baseline under 4 hazards.
2. Geometric Grounding	Linear head on Cosmos 64d: no gain. GPU MLP: +0.042 RFS. InternVLA: no gain (generation $\neq$ discrimination).
3. Causal Localization	Failure modes trace to scalar interactions, not object labels. 1.419 RFS oracle gap $\rightarrow$ visual-geometric bridge missing.
AlpaSim bridge	Same token-policy API runs through a front-camera/route/ego adapter with paired scene matrices and per-frame axis logs.
AlpaSim finding	10 paired WOD-E2E clips expose axis-specific transfer: clamping helps route geometry, veto helps progress, source decay helps lane behavior.
WOD-E2E champion	**7.845 RFS** (5-fold), +0.823 vs official Waymo baseline.

Conclusion & Next Steps

What we showed:

- **Discrete bottleneck:** 9 tokens, 0 collisions, 27× gauntlet gain.
- **Grounded visual signal:** Cosmos helps only with a nonlinear readout.
- **Bottleneck localized:** 1.419 RFS oracle gap points to the visual-to-geometric bridge.
- **AlpaSim bridge:** same policy API, second execution stack, paired WOD-E2E scenes, per-axis logs.

General principle:

Grounded action interfaces make transfer failures measurable instead of hidden inside one aggregate score.

Next steps:

1. **Preference-aligned visual encoder**
   Fine-tune on triplets (frame, win, loss) to close the oracle gap.
2. **Intent calibration**
   Stratified reweighting by intent; GO_RIGHT regret is 2.063.
3. **Test-set run**
   Pipeline complete; missing gated Waymo test TFRecords.
4. **Physical deployment path**
   Obstacle sensors + route command → Spotlight Reflex.

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`docs/simulation.md` · `docs/wod-e2e-system-walkthrough.md`
`docs/corl2027/paper.pdf` · `docs/corl2027/paper.txt`